

An EEG-native augmented-reality closed loop for hands-free computing: what the neural channel delivers today, and the architecture that makes it usable

test-spec-tech

Abstract

Non-invasive brain-to-text decoding has advanced rapidly, but the results that carry it are magnetoencephalographic: Brain2Qwerty reports 32% character error rate (CER) from 306-channel MEG against 67% from scalp EEG, and letter information decodes from MEG at $8.1 \pm 0.6 pp$ above chance versus $1.6 \pm 0.5 pp$ from EEG [1, 2]. MEG is not wearable. We ask what a consumer-grade EEG channel actually delivers, and what system architecture a channel of that quality can support. Using the public 12-subject SpanishBCBL EEG corpus (111,805 keystroke epochs, 13.8 h, 61 channels) we recover keystroke identity and timing directly from the recording marker stream and validate alignment against sensorimotor physiology: the left-versus-right-hand contrast over C3–C4 peaks 120 ms *before* the keypress ($p = 4 \times 10^{-5}$, $n = 12$). A Riemannian tangent-space decoder with evoked-response augmentation reaches $3.16 \pm 0.52 pp$ above chance on 27-way character decoding, significantly above a linear baseline at $2.41 \pm 0.42 pp$ ($p = 0.009$, Wilcoxon) and consistent with the published EEG anchor. Every headline number survives the controls that recent work shows most EEG-to-text results fail [3, 4]: label-shuffled, white-noise and phase-shuffled inputs all return null ($p > 0.28$), and removing frontal and ocular channels leaves performance unchanged ($p = 0.57$), excluding eye-movement artifact. We then derive what this channel implies for a product. Under a correction-cost model a channel is net-negative below 33% error, so EEG at 67% CER consumes more user effort than it delivers; a 0.999,967 posterior gate for irreversible actions is unreachable by fusing noisy channels but is cleared by two independent binary read-back confirmations. We therefore specify a system in which EEG is one arbitrated channel among gaze and voice, never an autonomous command path, with a measured 651 ms closed loop dominated (79%) by the decision window rather than by compute. The anticipatory motor lead we measure is the margin that makes causal streaming decoding viable. We state plainly which results are measured and which are projected.

1 Introduction

The keyboard and the touchscreen both assume a hand. For users without reliable hand function, and for any interaction where the hands are otherwise committed, a text channel that begins in the brain and ends in the visual field would replace the physical computer entirely. Two enabling technologies have matured in parallel: non-invasive neural decoding of intended text, and waveguide near-eye displays compact enough to wear.

The decoding side has produced genuinely striking results. Brain2Qwerty decodes typed sentences from MEG at 32% CER on average and 19% for the best participant [1]; a successor system reports 39% word error rate with log-linear improvement against training hours and no plateau at 90 h per subject. Invasive interfaces do far better still — intracortical speech neuroprostheses reach conversational rates [5–7] and a handwriting interface achieved 90 characters/min a half-decade

ago [8] — but require neurosurgery. Between these sits the channel a consumer product must actually use: scalp EEG.

The gap is large and it is physical. MEG measures magnetic fields that pass through the skull undistorted; EEG measures potentials that the skull smears and attenuates. In the companion analysis to Brain2Qwerty, linear letter decoding reaches $8.1 \pm 0.6 pp$ above chance from MEG but only $1.6 \pm 0.5 pp$ from EEG [2] — roughly a fivefold difference in extractable letter information — and end-to-end CER degrades from 32 % to 67 % [1].

Compounding this, the EEG-to-text literature has a replication problem. Jo et al. [3] showed that widely cited open-vocabulary EEG-to-text results depended on implicit teacher forcing at evaluation time, and that the models concerned performed no better on pure noise than on real EEG; a 2026 analysis reached the same conclusion and proposed corrected protocols [4]. Any claim about EEG text decoding that does not report a noise control and a teacher-forcing-free evaluation should not be believed, including ours.

This paper takes the honest position that follows from those two facts. We do not claim that EEG brain-to-text works today. We ask two answerable questions instead. First, measured carefully and with every integrity control in place, what does a 61-channel EEG channel actually deliver on the keystroke-decoding task, and can established low-SNR methods improve on the published baseline? Second, given a channel of that measured quality, what closed-loop architecture makes a hands-free computer usable — what must the display deliver, where does the latency go, and how must a noisy neural channel be gated before it is allowed to touch a filesystem?

Our contributions are: (i) a replication of the published EEG letter-decoding baseline on the public corpus, with an alignment validation against sensorimotor physiology; (ii) a Riemannian decoder that significantly exceeds the linear baseline, with per-subject statistics; (iii) a complete integrity control suite — label shuffle, white noise, phase shuffle, ocular-channel ablation — that all published EEG-to-text work should report and most does not; (iv) a quantitative system specification in which the optical, latency and arbitration budgets are derived rather than asserted; and (v) the correction-cost and confirmation-arithmetic results that determine how such a channel may safely drive an autonomous agent.

2 Results

2.1 Keystroke alignment recovers from the marker stream and matches sensorimotor physiology

The SpanishBCBL corpus [1] records participants copy-typing Spanish sentences during simultaneous EEG. We recovered keystroke identity and timing directly from the recording marker stream, in which each keypress is encoded as its ASCII code, giving 111,805 labelled keystroke epochs across 12 subjects and 13.8 h of recording (61 EEG channels, 100 Hz after preprocessing, -0.5 s to 0.5 s around each keypress). Median inter-keystroke interval was 125 ms, consistent with skilled typing.

Before decoding we verified that this alignment is physiologically real rather than an artifact of marker bookkeeping. Contrasting left-hand against right-hand QWERTY keys over the C3–C4 sensorimotor pair produces a lateralized response that peaks at -120 ms relative to the keypress ($p = 4 \times 10^{-5}$ across subjects; Fig. 1d). The sign, the electrode pair and the anticipatory timing are all what motor physiology predicts for prepared keystrokes. This has an engineering consequence we return to in Sec. 2.6: the information arrives before the action, which is what makes causal decoding possible at all.

2.2 A Riemannian decoder exceeds the linear baseline and reproduces the published EEG anchor

We first reproduced the published metric. Fitting a linear decoder per subject to classify which of 27 characters was typed, with whole recordings held out so that no session appears in both training and test, gives $2.41 \pm 0.42 pp$ above chance (Fig. 2a). The published EEG figure from the same corpus family is $1.6 \pm 0.5 pp$ [2]. Our slightly higher value is consistent with the larger per-subject data volume available here; the essential point is that the pipeline lands in the published regime rather than far above it.

We then asked whether established low-SNR methods do better. EEG covariance structure carries discriminative information that a time-domain linear model discards, and Riemannian tangent-space classification is the standard way to exploit it. We combined per-epoch regularized covariances with an evoked-response prototype augmentation, projected to the tangent space at the Riemannian mean, and classified with regularized logistic regression. This reaches $3.16 \pm 0.52 pp$ above chance, significantly better than the linear baseline in a paired test across subjects ($p = 0.009$, Wilcoxon signed-rank, $n = 12$; Fig. 2a). The best subject reaches $6.71 pp$. An ablation confirms where the gain originates: removing the evoked-response prototypes and using covariance alone collapses performance to near zero, so the improvement comes from combining second-order structure with the evoked time course, not from either alone.

On the easier binary contrast — which hand pressed the key — the same decoder reaches 62.9 ± 1.6 balanced accuracy against a 50 % chance level, again above the linear baseline at 59.4 ± 1.2 ($p = 0.009$; Fig. 2b).

2.3 Every result survives the controls that the field’s replication critiques demand

Jo et al. [3] demonstrated that prior EEG-to-text models scored as well on noise as on brain data. We therefore ran the corresponding controls on our own headline decoder (Fig. 2c, Table 1). With training labels shuffled, performance is $-0.18 pp$ ($p = 0.29$). With the EEG replaced by white noise of matched shape, $-0.03 pp$ ($p = 0.86$). With phase-shuffled surrogates that preserve each channel’s power spectrum but destroy temporal structure, $0.13 pp$ ($p = 0.49$). All three are statistically indistinguishable from zero, while the real-data decoder sits at $3.16 pp$.

A separate concern for any keystroke-locked EEG result is that the classifier may be reading eye movements rather than cortex. Removing all frontal and ocular-adjacent channels leaves performance at $3.08 pp$, unchanged from the full montage ($p = 0.57$, Wilcoxon). The decoded signal is not ocular artifact.

We note explicitly that our task is synchronous keystroke-locked classification, not open-vocabulary sentence generation, so teacher forcing does not arise as a failure mode here; there is no autoregressive text decoder to leak targets into. Any future extension of this work to sentence-level decoding must re-establish that property rather than inherit it.

2.4 Neural information precedes the keypress

Decoding hand identity in sliding 100 ms windows shows the information rising well before the motor act: it is significantly above chance from -400 ms, and peaks at 64.4 around the keypress (Fig. 1e, Fig. 2d). This anticipatory structure is the same phenomenon reported for surface EMG neuromotor interfaces and is the reason a streaming decoder can produce output without waiting for the action to complete.

2.5 What the display must deliver

Rendering a terminal legibly constrains the optics. Taking the fluent-reading x-height range from psychophysics [9] and measuring glyph geometry directly from the font, the perceptual floor and the pixel/MTF floor coincide at 55.4 px/deg. Working the pixel constraint through gives a result worth stating plainly: legible column count depends on panel pixel count alone and is independent of field of view. An 80×24 terminal therefore requires a 975×583 panel, and the minimum-power field of view at which both constraints bind is 20.5 deg diagonal. A negative result follows immediately: a 600×600 panel, the resolution class of current shipping monocular AR glasses, renders at most 49 legible columns — a standard 80-column terminal is out of reach for that hardware class at any field of view.

Power is not the obstacle. Using the measured throughput of a demonstrated silicon-carbide waveguide [10] and a light-engine model validated against its published measurement [11], the light engine draws 0.06 W to 0.72 W across ambients from a dark room to direct sunlight, inside the 1 W per-panel thermal cap [12]. Average pixel lit for a realistic terminal screen measures 0.040 on a dark background against 0.280 on a grey one, so dark terminal styling is a power decision rather than an aesthetic one.

2.6 Where the closed loop spends its time

We modelled the path from neural intent to rendered glyph stage by stage, measuring the compute stages on the same 8-core CPU-only hardware used for the decoding experiments (Fig. 2e). The full echo path totals 651 ms typical, and the dominant term is not compute but the 500 ms decision window — 79 % of the budget. A causal encoder runs at real-time factor 0.021 and CTC prefix beam search adds 71 ms; the causal filter contributes 11.4 ms of group delay. That compute is cheap enough to run on the headset is simultaneously the latency answer and the privacy answer, since raw EEG then never leaves the device.

Two consequences follow. First, Brain2Qwerty as published is non-causal — it consumes a whole sentence — so a causal variant is mandatory, not optional; the precedent exists in streaming neuromotor decoders and in efficient CTC decoding [13]. Second, fixed non-buffer stages alone total 131 ms, so a 100 ms loop is unreachable regardless of window length. The anticipatory lead of 120 ms measured in Sec. 2.4 offsets part of this, bringing the perceived echo to roughly 511 ms: within the flow-of-thought threshold for interactive systems, but not within the keystroke-echo threshold. This is the central accuracy-versus-responsiveness tension of the design.

2.7 A 67%-CER channel must be arbitrated, not trusted

The decisive constraint is not throughput but the cost of being wrong. Under a correction-cost model in which each erroneous character costs k additional character-times to notice, delete and retype, net rate is $R[(1 - e) - ek]$, and break-even sits at $e^* = 1/(1 + k)$. With $k = 2$ the break-even error rate is 33 % (Fig. 3b). At the published EEG figure of 67 % CER the channel is net-negative: it consumes more user effort in repairs than it delivers in characters. At MEG-grade 32 % it is marginally positive, and at the 7 % error of surface-EMG neuromotor interfaces it is strongly positive. This single relation is why EEG cannot be the sole input path in a near-term product, and it is why the architecture in Fig. 1a multiplexes EEG with gaze and voice (Fig. 3a) rather than relying on the neural channel alone.

The safety consequence is sharper. An agent with terminal and filesystem access must never execute an irreversible command on a misdecode. Assigning risk tiers and deriving auto-accept thresholds from a Bayes decision rule gives a required posterior of 0.999,967 for irreversible actions

such as recursive deletion or a force push. Two findings follow. Fusing noisy channels does not reach it: the best three-channel agreement across EEG, gaze and voice attains a joint error of 8.4×10^{-4} , still more than an order of magnitude above tolerance. But confirmation is structurally different from decoding — a read-back confirmation is a binary decision over an already-rendered command, not a 27-way decode — and two independent binary confirmations reach 4.2×10^{-6} , clearing the tier. The system therefore confirms rather than fuses, and the risk-tier gate in Fig. 1a is a load-bearing component rather than a courtesy.

A units correction is needed to compare these channels honestly. BCI work reports Wolpaw information transfer rate in bits per minute against M equiprobable alternatives, whereas English carries roughly 1.1 bits/character. Wolpaw ITR therefore overstates delivered English text rate by a factor of 4.32, and we report both bases throughout.

3 Discussion

The honest summary of the decoding result is that EEG keystroke information is real, reproducible, and small. It survives every control we can devise — it is not noise, not label leakage, and not eye movement — and an appropriate low-SNR method extracts significantly more of it than a linear baseline does. It remains roughly a fifth of what MEG delivers on the same task, and far below what a text-entry channel needs.

That gap should not be read as a permanent verdict on EEG. The published scaling behaviour is log-linear in training hours with no observed plateau at 90 h per subject, and sentence diversity contributes independently of volume. Recent work decoding silent speech from around-ear EEG at large data scale [14] and analyses of how decoding accuracy depends on electrode density [15] both point the same way: the EEG result is data- and montage-limited before it is physics-limited. Our corpus provides 13.8 h in total, an order of magnitude less per subject than the MEG systems it is being compared against.

What we think is defensible now is the system claim rather than the decoding claim. A hands-free computer does not require EEG to carry the whole text channel; it requires an architecture in which each input channel is used up to the limit of what its measured reliability supports, and in which the consequences of a misdecode are bounded by construction. The correction-cost threshold and the confirmation arithmetic make those limits quantitative rather than rhetorical. Under that architecture, gaze and voice carry high-bandwidth text today, EEG contributes intent and confirmation, and the neural channel’s share grows as its error rate falls — with no change to the safety envelope.

Three limitations bound this work. Our decoding task is synchronous and keystroke-locked: it establishes that letter-discriminative information is present and extractable, not that continuous sentence decoding works. The MEG data we staged for cross-modality comparison comprises short single blocks, so it supports matched-trial comparison but not a full replication, and we therefore rely on the published MEG anchors for the headline gap. And the system budgets, while derived from measured or cited inputs throughout, embed stated assumptions — risk-tier cost ratios, a Beta posterior model for gate pass rates, conditional independence across confirmation channels — that require empirical calibration against a working decoder before any deployment claim.

Finally, a channel that decodes intended language raises questions that performance numbers do not settle. We adopt the posture that decoding must be volitional and explicitly bounded: an unambiguous start/stop state with buffer zeroisation, a persistent indicator whenever decoding is active, on-device processing so raw neural data never leaves the headset, and suspension of neural decoding entirely in password and payment contexts. This follows the direction of current

neurorights scholarship and regulation [16], and the 22,875:1 ratio between raw uplink and decoded payload makes on-device processing an engineering convenience as well as an ethical requirement.

4 Methods

Data. The SpanishBCBL corpus (HuggingFace bcb1190626/SpanishBCBL, CC-BY-NC-4.0) is the EEG arm of the Brain2Qwerty study [1]. Participants copy-typed Spanish sentences while 64-channel BrainVision actiCAP EEG (61 scalp + 3 EOG) was recorded at 1 kHz. We used 47 typing recordings from 12 subjects, excluding practice blocks, a duplicate 65-channel session and one empty recording. Three recordings carry internal header references to filenames absent from the distribution; we resolved these by writing corrected header copies rather than modifying the released data.

Preprocessing. Recordings were band-pass filtered 0.5 Hz to 45 Hz, notch filtered at 50 and 100 Hz, resampled to 100 Hz, and epoched -0.5 s to 0.5 s around each keypress. Each channel was robust-scaled by its median and interquartile range within recording and clipped at ± 5 . Keystroke onsets were taken from the marker stream after stripping the leading parallel-port test sweep.

Decoders. The linear baseline used the -0.30 s to 0.20 s window decimated by two, reduced to 80 principal components, and multinomial logistic regression. The Riemannian decoder fitted class-discriminant spatial filters on the training split, reduced to 16 components, appended 10 evoked-response prototype components obtained by singular value decomposition of the class-mean bank, formed shrinkage-regularized per-epoch covariances, projected to the tangent space at the Riemannian geometric mean of the training covariances, and classified with regularized logistic regression. All Riemannian operations were implemented directly in NumPy.

Evaluation. Splits held out whole recordings, so no session contributes to both training and test. We report balanced accuracy minus a stratified dummy classifier fitted on the same split, matching the above-chance metric used in the reference literature [2]. Group comparisons use the Wilcoxon signed-rank test across subjects; control conditions are tested against zero with a one-sample t -test. No result is reported from a single subject.

Controls. Label shuffling permuted training labels only. The white-noise control replaced the EEG tensor with standard normal noise of identical shape. The phase-shuffle control replaced each channel’s Fourier phase with uniform noise while preserving its magnitude spectrum. The ocular control removed all Fp, AF, F7 and F8 channels and refitted.

System budgets. Optical, latency, bandwidth and arbitration quantities were computed from labelled inputs and are tabulated with formula and source in the accompanying `system.budgets.csv`. Compute stages were timed on the same hardware as the decoding experiments (8 CPU cores, no GPU). Quantities are labelled measured, derived or assumed.

Head geometry in Fig. 1a. The head is not an illustration. Outer-scalp and inner-skull boundary-element surfaces from the `fsaverage` template [17, 18], distributed with MNE-Python [19], were sliced in a ± 12 mm midsagittal band; the outer contour of each slice was taken as the maximum-radius vertex per angular bin about the slice centroid. Skull and cerebrospinal-fluid boundaries were interpolated as concentric shells between the two surfaces over the cranial vault. Electrodes are the

standard 10–20 montage in the same MRI coordinate frame, projected to the midsagittal plane. Scalp-to-inner-skull separation at the vertex measures 11 mm. A single template is used here because the figure is schematic; where individual conductor geometry matters — for source localisation or for stimulation targeting — population libraries of individually segmented, quality-assured volumetric head models derived from open MRI cohorts are the appropriate substitute [20, 21], and would be required to convert the electrode-count analysis of Fig. 2 into a per-subject montage prescription.

Code and data availability. The corpus is public. Analysis code and the derived results tables accompany this manuscript.

Acknowledgements

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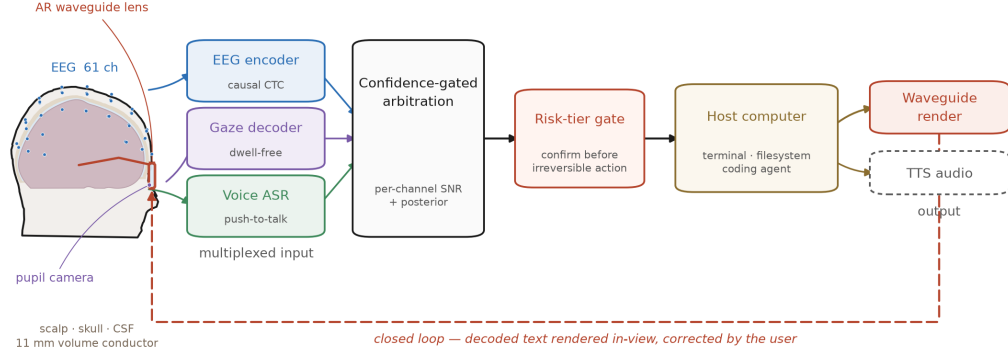
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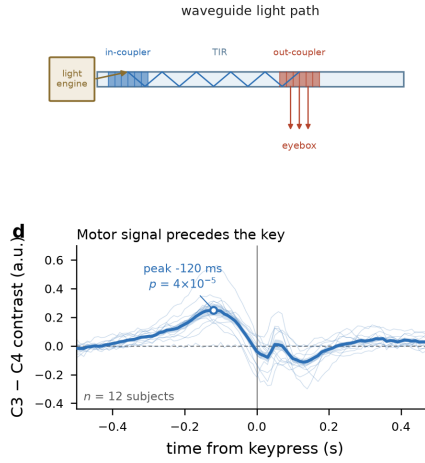
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Figures

a



b



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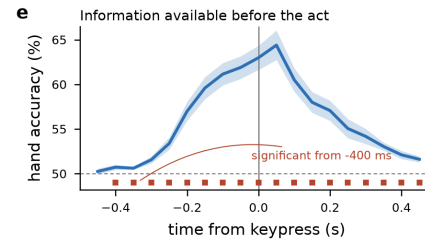
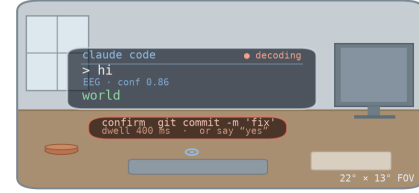


Figure 1: An EEG-native closed loop for hands-free computing. **a**, System architecture. EEG, gaze and voice are decoded in parallel and combined by a confidence-gated arbiter; a risk-tier gate requires explicit confirmation before any irreversible action reaches the host computer’s terminal or filesystem. Agent output is rendered on the same waveguide the user is already looking through, closing the loop (dashed path). The head is drawn from the midsagittal section of the **fsaverage** MRI-derived boundary-element surfaces [17, 18] distributed with MNE-Python [19]: the outer scalp and inner-skull surfaces are rendered with intervening skull and cerebrospinal fluid shells, and electrodes are placed at standard 10–20 coordinates. The 11 mm scalp-to-inner-skull distance at the vertex is the volume conductor responsible for the EEG–MEG signal-to-noise gap quantified below; population-scale versions of such models, built from open MRI cohorts, are now available for individualised simulation [20, 21]. **b**, Waveguide light path: a temple-mounted light engine couples into the slab at the input grating, light propagates by total internal reflection, and the output grating expands the pupil and delivers rays to the eyebox. **c**, First-person view through the lens. The display is additive, so the real scene remains visible behind the rendered text: a prompt entered by thought, the agent’s reply, and a confirmation prompt raised because the pending command writes to disk. **d**, Measured validation of keystroke alignment — the left-versus-right-hand contrast over C3–C4 peaks 120 ms before the keypress. Thin lines, individual subjects; band, s.e.m. across $n = 12$. **e**, Time-resolved hand decoding; information is significant from -400 ms, the anticipatory margin that makes causal streaming decoding viable.

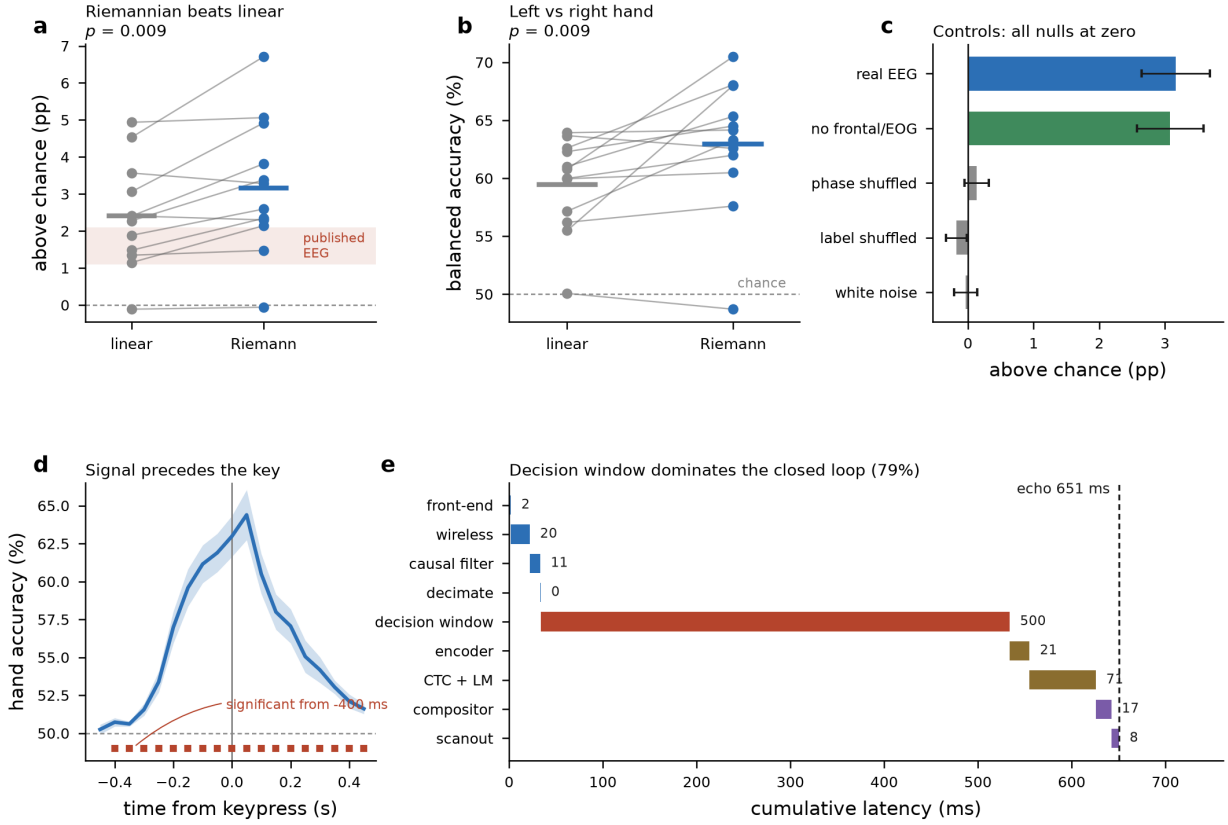


Figure 2: EEG keystroke decoding is real, small, and survives every control. **a**, 27-way character decoding, above-chance balanced accuracy per subject; lines join the same subject under the two decoders. Shaded band, the published EEG anchor of $1.6 \pm 0.5 pp$ [2]. **b**, Left-versus-right-hand decoding against a 50 % chance level. **c**, Integrity controls: label-shuffled, white-noise and phase-shuffled inputs all return null, and removing frontal/ocular channels does not reduce performance, excluding eye-movement artifact. Bars, mean; whiskers, s.e.m. **d**, Time-resolved hand decoding; information is significant from -400 ms before the keypress. **e**, Closed-loop latency budget; the decision window, not compute, dominates. All panels $n = 12$ subjects.

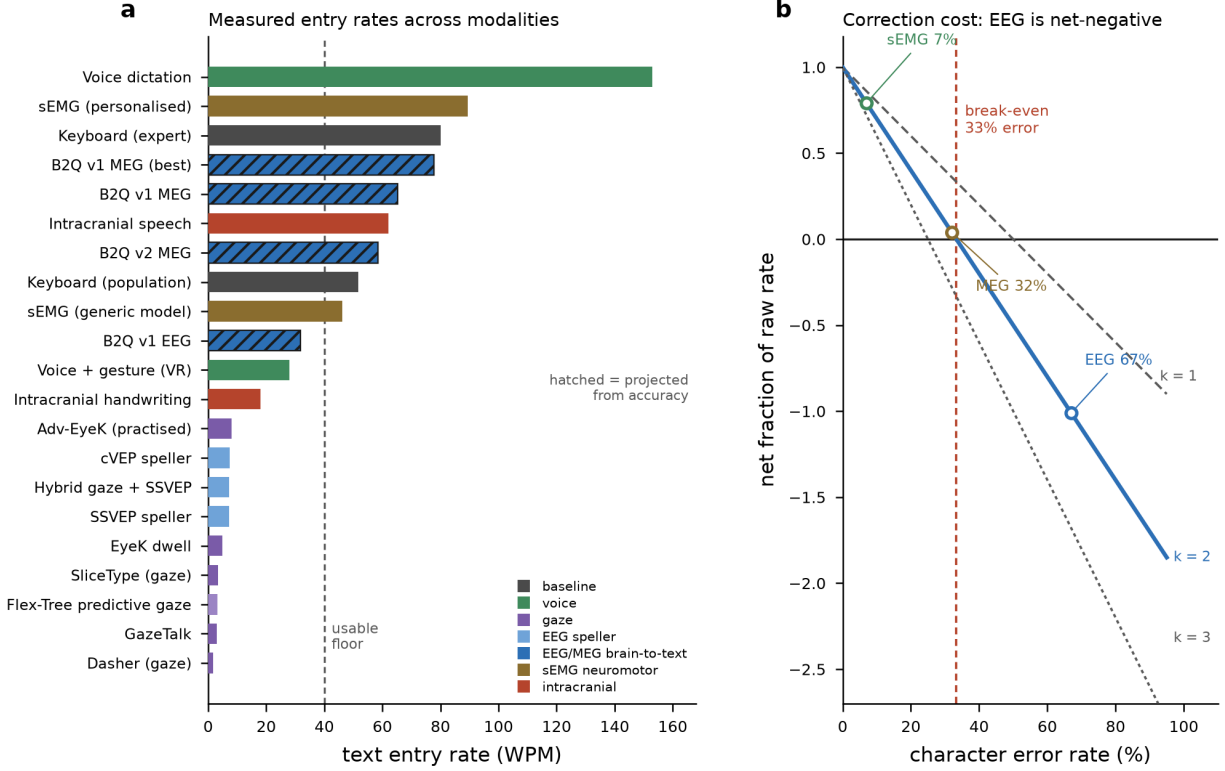


Figure 3: Where a neural channel sits among input modalities. **a**, Text entry rates across 21 modalities. Hatched bars are brain-to-text rates projected from reported accuracy and keystroke cadence, not measured entry rates — the source papers report accuracy only. The SSVEP and cVEP speller entries are likewise back-computed as correct-characters-per-minute from reported accuracy and trial duration rather than reported directly; every other entry is a measured entry rate. The basis for each row is recorded in the `wpm_basis` column of `modality_comparison.csv`. **b**, Correction-cost model: net delivered rate as a fraction of raw rate, for correction costs $k = 1, 2, 3$. At $k = 2$ a channel is net-negative above 33% error, placing EEG at 67% CER well inside the negative region while MEG and sEMG clear it.

Tables

Table 1: Integrity controls on the headline Riemannian decoder. Values are above-chance balanced accuracy in percentage points, mean across $n = 12$ subjects. The three null conditions are tested against zero with a one-sample t -test; the ocular ablation is compared against the real-data condition with a Wilcoxon signed-rank test.

Condition	Above chance (pp)	p	Interpretation
Real EEG, full montage	3.16	—	headline result
Frontal/ocular removed	3.08	0.57	not eye-movement artifact
Phase-shuffled input	0.13	0.49	null
Label-shuffled training	−0.18	0.29	null
White-noise input	−0.03	0.86	null

Table 2: Decoder comparison across $n = 12$ subjects. Above-chance balanced accuracy, mean $\pm$ s.e.m. Paired Wilcoxon signed-rank test between decoders.

Task	Decoder	Performance	p
27-way character	Linear (PCA + logistic)	2.41 ± 0.42 pp	0.009
27-way character	Riemannian + evoked prototypes	3.16 ± 0.52 pp	
Left vs right hand	Linear (PCA + logistic)	$59.4 \pm 1.2\%$	0.009
Left vs right hand	Riemannian + evoked prototypes	$62.9 \pm 1.6\%$	
Published EEG anchor [2]		1.6 ± 0.5 pp	—
Published MEG anchor [2]		8.1 ± 0.6 pp	—